

# 05: Conditions

Stat 230 - Spring 2026

## Note

This is the quarto version of the R activity, designed for opening in maize/RStudio. If you prefer to use the tutorial version, you can use <https://aluby.github.io/stat230/activities/05-conditions-tutorial>

```
library(tidyverse)
library(ggformula)
theme_set(theme_minimal())
```

## 1 Example 1: Making residual plots in R

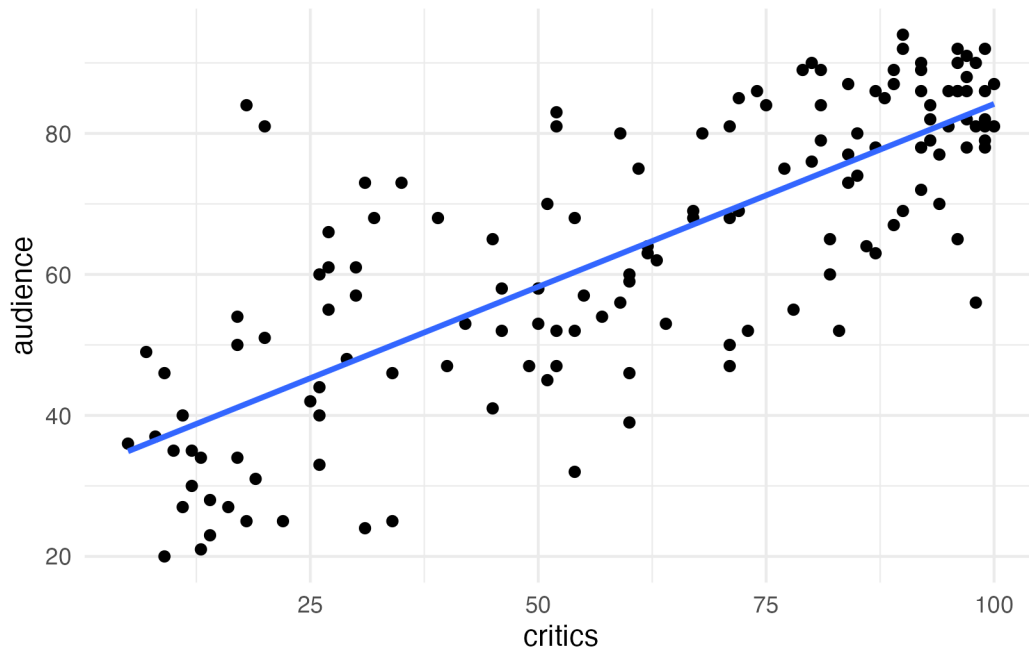
Recall the movies dataset from the R activity we did on Day 02. (`head()` prints out the first few rows)

```
movies <- read.csv("https://aloy.github.io/stat230-materials/data/movie_scores.csv")
head(movies)
```

|   | film                    | year | critics | audience |
|---|-------------------------|------|---------|----------|
| 1 | Avengers: Age of Ultron | 2015 | 74      | 86       |
| 2 | Cinderella              | 2015 | 85      | 80       |
| 3 | Ant-Man                 | 2015 | 80      | 90       |
| 4 | Do You Believe?         | 2015 | 18      | 84       |
| 5 | Hot Tub Time Machine 2  | 2015 | 14      | 28       |
| 6 | The Water Diviner       | 2015 | 63      | 62       |

On that day, we made the following scatterplot:

```
gf_point(audience ~ critics, data = movies) |>
  gf_lm()
```



and fit this regression model:

```
movie_lm <- lm(audience ~ critics, data = movies)
movie_lm
```

Call:

```
lm(formula = audience ~ critics, data = movies)
```

Coefficients:

| (Intercept) | critics |
|-------------|---------|
| 32.3155     | 0.5187  |

## 1.1 Finding residuals

In order to make residual plots, we first need to find the residuals. While we can do this by hand for a couple of points, it would be extremely tedious to do it for *every single data point*! Instead,

we'll use the `augment()` function from the `{broom}` R package to create an augmented movie dataset. This takes the original X and Y variables and adds the  $\hat{y}$  values (`.fitted`) and residuals (`.resid`) (along with some other columns that we'll use later)

Run the following code chunk and take a peek at the `movie_aug` dataset. Can you find the:

- original X variable?
- original Y variable?
- Predictions?
- Residuals?

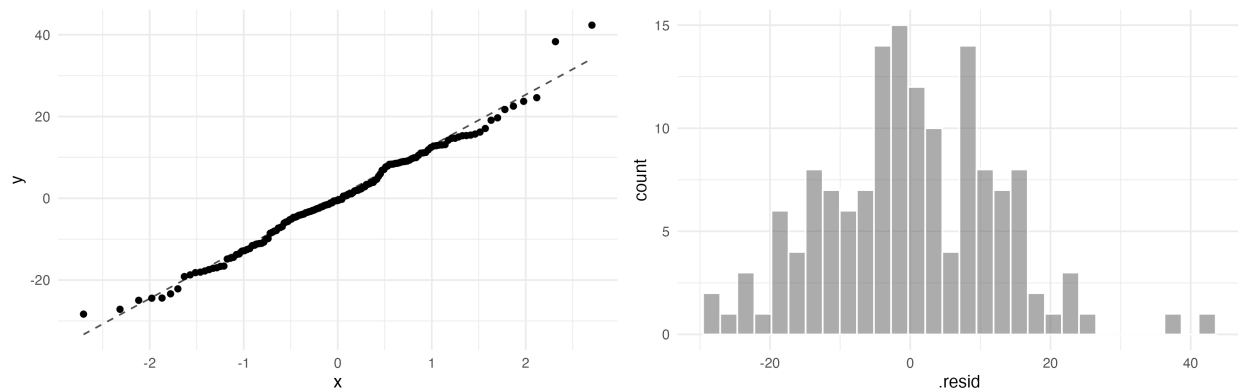
```
library(broom)
movie_aug <- augment(movie_lm)
movie_aug
```

```
# A tibble: 146 x 8
  audience critics .fitted .resid   .hat .sigma   .cooksd .std.resid
    <int>   <int>   <dbl> <dbl>   <dbl> <dbl>   <dbl>   <dbl>
1      86     74    70.7  15.3  0.00816  12.5  0.00618    1.23
2      80     85    76.4   3.60  0.0113   12.6  0.000474    0.289
3      90     80    73.8  16.2  0.00963  12.5  0.00818    1.30
4      84     18    41.7  42.3  0.0208   12.1  0.123     3.41
5      28     14    39.6 -11.6  0.0235   12.5  0.0105   -0.934
6      62     63    65.0  -2.99  0.00688  12.6  0.000199   -0.239
7      53     42    54.1  -1.10  0.00954  12.6  0.0000374  -0.0882
8      64     86    76.9 -12.9  0.0116   12.5  0.00633    -1.04
9      82     99    83.7  -1.66  0.0179   12.6  0.000163   -0.134
10     87     89    78.5   8.52  0.0129   12.6  0.00305     0.684
# i 136 more rows
```

## 1.2 Making residual plots

The code below makes a normal QQ plot and a histogram of the residuals. Do you think the normality assumption is satisfied?

```
gf_qq(~.resid, data = movie_aug) |>
  gf_qqline()
gf_histogram(~.resid, data = movie_aug, col = "white")
```



### 1.2.1 Checkpoint question

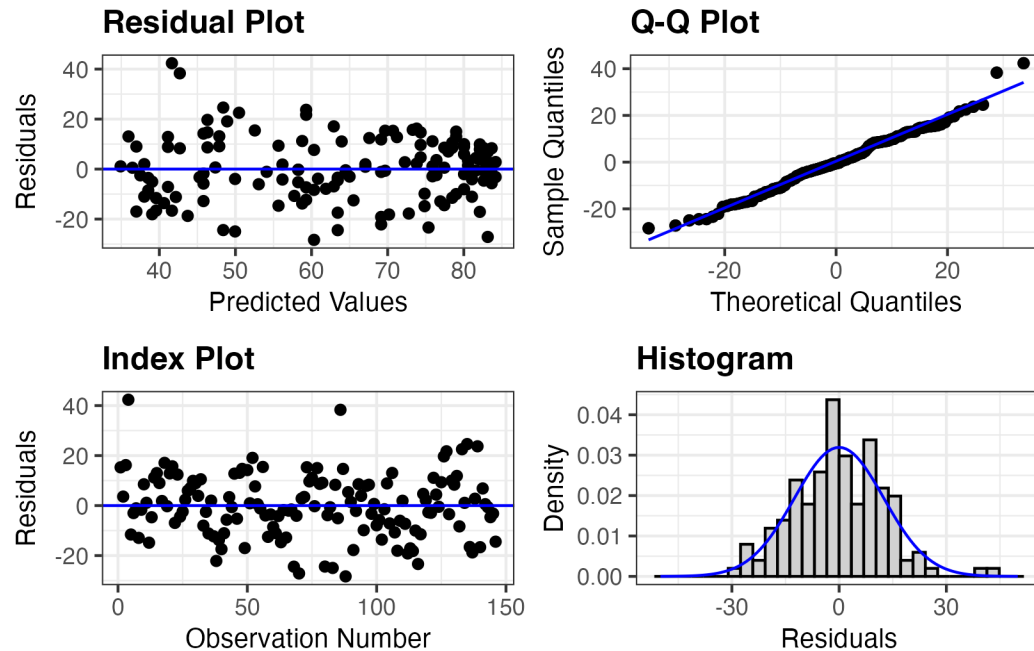
Use the `movie_aug` dataset to make a scatterplot with `.fitted` on the x-axis and `.resid` on the y-axis. List the SLR conditions you can assess with this plot, and whether or not they appear to be satisfied.

```
# your code here
```

### 1.3 A shortcut

We will make a **ton** of residual plots this term. A shortcut is to use the `resid_panel` function within the `{ggResidpanel}` package:

```
library(ggResidpanel)
resid_panel(movie_lm)
```



### 1.3.1 Checkpoint question

Which panel(s) would you look at for each of the following SLR conditions. List as many panels as possible, along with how you would identify any violations:

- Linearity:
- Independence:
- Normality:
- Equal variance:

## 2 Example 2: $R^2$ in R

### 2.0.1 Checkpoint question

Find the  $R^2$  value for the `movie_lm` model using `summary()`. Confirm that it is equal to  $r^2$

```
# your code here
```

## 2.1 Summary

### 2.1.1 Checkpoint question

Suppose that we fit a linear regression model to a sample of 1,000 observations and we find evidence in the normal Q-Q plot of a violation of the normality assumption, but no evidence in the residual plot of violations of the linearity and constant variance assumptions and no reason to suspect a violation of the independence assumption. Which of the following inferences are valid, and which are not?:

- p-value from the hypothesis test  $H_0 : \beta_1 = 0$  vs  $H_A : \beta_1 \neq 0$
- Confidence interval for  $\beta_1$
- Confidence interval for the mean of Y at some X value
- Prediction interval for Y at some X value

## 2.2 Function quick reference

The following table summarizes the functions we learned today:

| Function                            | Purpose                                                          |
|-------------------------------------|------------------------------------------------------------------|
| <code>augment(model)</code>         | Create augmented dataset that includes predictions and residuals |
| <code>gf_qq(~variable, data)</code> | Make a qq plot                                                   |
| <code>gf_qq_line()</code>           | Overlay a straight line on the qq plot                           |
| <code>resid_panel(model)</code>     | Shortcut for making a panel of residual plots                    |